

Caliber

A transparent, AI-powered US immigration platform. It reads your real achievements the way a USCIS officer would — telling you which visas you qualify for and why, tracking the rules that affect you, and walking you from a student visa to a green card.

TRACK	BUILD WINDOW	STATUS	NAME
Best Immigration + Best Overall	~5 days · solo full-stack / AI	Design brief v2	Caliber (provisional)

01 In one breath

People with real, world-class achievements routinely don't know they qualify for an O-1 "extraordinary ability" visa — and a lawyer's eligibility assessment costs **\$1,000–\$2,000** and takes a week.

Caliber lets you paste your public profile — CV, publications, GitHub, a few press links — and in about a minute returns a grounded verdict: which of the legal criteria you meet, **why**, where your gaps are and how to close them, and a first-draft petition letter. Every judgment cites the exact USCIS rule it used. It covers a whole family of achievement-based visas, tracks live rule changes, and gives F-1 and H-1B holders a path forward.

Who it's for

Foreign-born researchers, engineers, founders and creatives in the US (or hoping to be) — from F-1 students to O-1/green-card candidates who can't afford to find out where they stand.

Why it wins

Real multi-agent AI doing a lawyer's judgment work, grounded in real law, honest about its limits, in a polished end-to-end product with a platform story.

The one-line pitch

"Caliber reads your achievements the way a US immigration officer would, argues both sides of each legal criterion, and hands you a transparent scorecard plus a draft petition — turning a \$2,000, week-long lawyer assessment into a free, 60-second one, for a whole family of visas."

02 The problem & who it affects

The O-1 and EB-1A are US pathways for people of "extraordinary ability." Unlike the H-1B, they have **no lottery and no annual cap** — but they're poorly understood, so thousands of genuinely qualified people never pursue them. Meanwhile F-1 students and H-1B workers face a maze of deadlines and odds with no clear map.

Three real frictions

You don't know if you qualify

Eligibility hinges on meeting a set of dense legal criteria most people can't map to their own résumé.

Finding out is expensive

A lawyer's assessment is \$1,000–\$2,000 and days of back-and-forth, just to learn *whether* it's worth filing.

The advice is opaque

You get a yes/no, rarely the *why*, and almost never a concrete plan to strengthen a weak case.

Who this hits

International students on the H-1B cliff, foreign-trained researchers, immigrant founders — exactly the Open Atlas audience.

The gap in the market: today you get either a *free static quiz* that just asks you to self-rate, or an *expensive opaque service*. Nothing does the actual reasoning, transparently, for free — or connects the visas into one journey.

03 The full product

One platform, four connected areas. Landing → Google sign-in → dashboard, with:

1 Assessment engine

Paste your profile → adversarial agent panel → scorecard (met criteria, why, gaps, draft letter, PDF). Covers a family of achievement-based visas — see §05.

2 F-1 / H-1B pathway

A lighter, rules-based tool: H-1B lottery odds, F-1 STEM-OPT eligibility, status timeline — and a bridge into the assessment engine. See §09.

3 Policy Watch

Live feed of recent immigration rule changes, AI-summarized and classified by visa (incl. F-1 & H-1B), with personalized alerts. See §10.

4 Saved history

Your past assessments and alerts, tied to your Google account — the reason to sign in and come back.

The platform story: we meet you as an F-1 student, show you your H-1B odds and their limits, then walk you to an O-1 or a green card — tracking the rules that change along the way. That's a journey, not a quiz.

04 How the assessment works — simple version



1 • You land & sign in. One line explains it; you sign in with Google so assessments are saved.

2 • You pick a visa. A selector — O-1A, EB-1A, O-1B, and more. The same engine drives each; only the rule set changes.

3 • You give it your stuff. Paste links you already have — publications, GitHub, a company page, a couple of news articles — and optionally drag in your résumé PDF. Add a couple of facts a link can't tell it (field, salary range). Hit *Analyze*.

4 • It works, and you watch it think. "Reading your profile... mapping evidence... checking the USCIS standard..." — visible reasoning, not a canned form.

5 • The scorecard. Verdict up top (e.g. *4 of 8 — likely eligible*). Each criterion shows Met / Partial / Gap, a one-line reason, and a link to the exact USCIS rule behind it.

6 • You act & leave with something. *Draft petition letter* writes the argument from your Met criteria. See *gap runway* gives each gap a concrete plan. Download the PDF — the exact thing you'd hand a real lawyer, minus their \$2,000 fee.

05 Visa coverage — two kinds of tool

Caliber's scorecard engine answers one shape of question: "*do your achievements meet N of M legal criteria?*" Some visas are that shape; F-1 and H-1B are not — so they get a different, rules-based tool (§09). Nothing is shoehorned.

Assessment engine — evidence-scored visas

Same engine, one config pack (criteria list + threshold + policy text) per visa.

VISA	WHAT IT'S FOR	RULE	SCOPE
O-1A	Extraordinary ability — sciences, business, athletics	3 of 8	Hero · deep
EB-1A	Extraordinary-ability green card	3 of 10	Demo pack
O-1B	Extraordinary ability — arts	3 of 6	Stretch pack
EB-1B	Outstanding professor / researcher	2 of 6	Stretch pack
EB-2 NIW	National interest waiver	3-prong test	Stretch pack

Pathway module — F-1 & H-1B

Not evidence-scored — these are rules, odds, and timelines, plus a bridge into the engine above.

VISA	WHAT IT'S FOR	TOOL SHAPE	SCOPE
H-1B	Specialty-occupation worker (lottery, capped)	Odds + eligibility + pathway	Lean module
F-1	Student (+ OPT / STEM OPT)	Eligibility + status timeline	Lean module

Why include F-1/H-1B at all: they're the largest slice of the audience and the natural *on-ramp*. The bridge funnels them into the O-1/EB-1A engine — turning "an O-1 checker" into "a company that owns the immigrant's whole journey."

06 What makes it different

The category isn't empty — and the judges know it. Caliber's edge isn't "first," it's **mechanism + honesty**.

WHAT EXISTS TODAY	WHAT IT DOES	CALIBER'S DIFFERENCE
Law-firm eligibility quizzes	Ask you to self-rate; funnel to a paid service	We read your real evidence and decide the mapping for you
AI petition drafters (paid)	Generate documents behind a black box	Every score cites the real USCIS rule — transparent, not opaque
ChatGPT "do I qualify?"	Plausible, ungrounded, uncited guess	Grounded retrieval + adversarial checking + deterministic verdict

The four defensible wedges

1 • Evidence → criterion mapping

The AI decides which criterion each achievement supports and how strongly — the lawyer's judgment work, not a self-report form.

2 • Grounded & cited

Every verdict links to the specific USCIS Policy Manual passage it reasoned from. No invented law.

3 • Gap runway, not yes/no

The valuable output is the plan to close a weak case — nobody self-serve does this.

4 • The RFE predictor

An "Examiner" agent argues how an officer would push back *before you file* — a genuinely novel output.

07 The core principle: AI proposes, code disposes

This is the design decision that makes Caliber accurate enough to trust — and proves it's a system, not an API key with a UI.

The rule: the LLM **extracts evidence and argues** each criterion. Deterministic code holds the legal rules, does **all counting and math**, and renders the **final verdict**. Retrieval guarantees every claim points to real law. The AI never decides the outcome — it informs a rule that does.

Deterministic — code / rules / data

- **The verdict itself** — counting "≥N of M met" is a rule, not an opinion.
- **Criteria & thresholds** — hardcoded per visa from the regulations.
- **Citation grounding** — the AI can only cite policy passages that exist in the store.
- **H-1B odds & salary percentile** — real math on real published data.
- **Counts** — citations, stars, publications fetched from APIs, never estimated.
- **STEM-OPT check** — lookup against the DHS designated-degree list.

AI-handled — the right job for an LLM

- **Extraction** — messy PDF/HTML → structured evidence.
- **Mapping** — which criterion a given achievement supports.
- **Per-criterion reasoning** — the Advocate / Examiner arguments.
- **Gap advice** — contextual, field-specific runway.
- **Drafting** — Met criteria + evidence → petition prose.
- **Policy summaries** — dense rules → plain language + "affects you."

Why this answers "aren't you just wrapping an API?" The decision layer is ours. The model is a bounded extractor/reasoner inside guardrails it can't override.

08 The multi-agent architecture

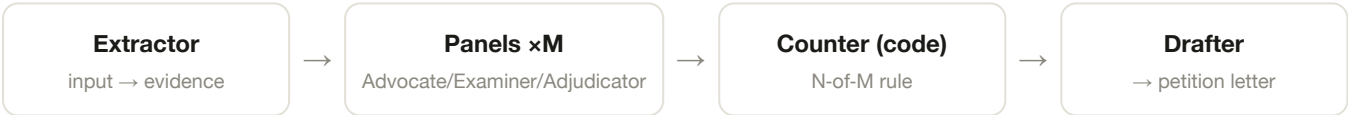
US visa adjudication is **inherently adversarial**: a lawyer files the strongest case, an officer looks for reasons to deny (a "Request for Evidence"). Caliber mirrors that with a per-criterion panel — so the structure reflects reality, not novelty for its own sake.

The panel — run per criterion

AGENT	ROLE	REAL-WORLD ANALOG
Advocate	Argues the evidence <i>meets</i> this criterion, citing the strongest facts	The petitioner's lawyer
Examiner	Argues why USCIS would <i>reject</i> it — surfaces the likely RFE	The adjudicating officer
Adjudicator	Weights both against retrieved policy → Met / Partial / Gap + confidence + reasoning	The decision

Then code takes over: it counts the Met criteria and applies the N-of-M rule. The panel judges each criterion; **it never decides the overall verdict.**

The full graph



Efficiency (deliberate): the full 3-agent panel only spins up on *borderline* criteria — a slam-dunk Met or an obvious Gap resolves in a single pass. Cheaper, faster, and it's exactly the contested criteria where the two-sided argument is worth showing on camera.

Standout feature: the Examiner's objections *are* an RFE predictor — "here's how an officer would push back, before you file." That's the 20-second clip that carries the demo video.

09 The F-1 / H-1B pathway module

A different, lighter tool — mostly deterministic rules and published data, AI only for the plain-language explanation and the "which path next" reasoning.

H-1B

TOOL	HOW
Lottery odds calculator	Real math on USCIS's published registration + selection counts per cap season
Specialty-occupation / degree match	Rules + AI reading a user-provided job description vs. your degree
Cap-exempt check	University / nonprofit? Rule-based

F-1

TOOL	HOW
STEM-OPT eligibility	Lookup against the DHS designated STEM-degree list (CIP code)
OPT / status timeline	Rules-based tree — 12mo OPT, +24mo STEM, grace periods, unemployment limits
The H-1B cliff planner	"OPT runs out here → your options are..."

The bridge (the strategic payoff): *"Your H-1B lottery odds are ~26%. Here's your backup: based on your profile you may already meet 3 of the O-1A criteria →"* — one click into the assessment engine. This is why F-1/H-1B earn their place: they funnel the biggest audience into the strongest asset.

10 Policy Watch — living rule tracking

A feed of recent US immigration changes, AI-summarized and classified by visa — **including F-1 and H-1B**, which affect the widest audience. This is what turns Caliber from a one-time quiz into a product you return to.

Each update card shows

[what](#) changed · [who](#) is affected · [visa](#) tags · [effective](#) date · plain-language summary · link to the source rule.

Where the AI does real work

Summarize

Dense Federal Register legalese
→ one clear sentence.

Classify

Tag each change to visa types +
an impact level.

Personalize

Cross-reference a change
against the user's saved
assessment: "affects criterion 3
on your scorecard."

Deterministic vs AI: code fetches the Federal Register API, date-filters, dedups, and pulls effective dates from structured fields; the AI summarizes, classifies, and reasons about personal impact.

Honest caveat (stated in the demo): major rule changes don't happen every month for every visa. The feed shows **real** Federal Register items — volume varies, and we never fabricate a change. F-1 and H-1B churn most, so they anchor the feed.

11 Data sources — only what's genuinely available online

Every source below is reachable with a free, online API or a public dataset. Anything that isn't (e.g. LinkedIn, which has no usable API and blocks scraping) has been **cut from scope** — nothing on the critical path depends on a source that will fail on demo day.

DATA NEED	SOURCE	STATUS
Criteria + thresholds (all visas)	8 CFR 214.2 / 204.5 — public domain	Available
USCIS Policy Manual text (grounding)	uscis.gov — public domain	Available
Publications + citation counts	OpenAlex API — free, no key	Available
Code impact (repos, stars)	GitHub REST API — free w/ token	Available
Résumé / CV	User-uploaded PDF → parsed	Available
Press / media coverage	User-pasted article links → AI evaluates	Available
H-1B lottery odds	USCIS cap-season registration/selection stats	Available
F-1 STEM-OPT eligibility	DHS designated STEM-degree list	Available
Rule-change feed (Policy Watch)	Federal Register API — free JSON	Available
Salary percentile (top-10% check)	BLS OES public wage dataset	Available, sampled
LinkedIn import	No usable API	Cut

On the sampled item: BLS wage data is free and online but bulky. The demo loads a real BLS snapshot for common occupations; full coverage is a post-hackathon plumbing task — stated openly, not hidden.

12 Tech stack

Lean, one language where possible, free at every layer. Fewest moving parts wins a 5-day solo build.

LAYER	CHOICE	WHY
Framework	Next.js (App Router) + TypeScript	UI + API in one app, one deploy
UI	Tailwind + shadcn/ui	The clean product look, fast
Auth	Auth.js (NextAuth) — Google OAuth	Real sign-in + saved per-user history
LLM	Google Gemini (2.5 Flash) — free tier	Free, native JSON output, 1M-token context fits the policy manual
Agent orchestration	Vercel AI SDK / thin TS loop	Runs the panel graph in-app, no Python service
Knowledge / retrieval	Curated policy chunks + embeddings	Criterion→law is a fixed lookup; semantic search for gap advice
Data fetchers	GitHub, OpenAlex, Federal Register APIs	Real deterministic data
Storage	Postgres (Neon/Supabase free tier)	Users, assessments, cached updates
PDF export	Browser print-to-PDF (CSS @media print)	No PDF library needed
Deploy	Vercel	Instant, free, live demo URL

Optional bonus prize: the agent graph is a natural fit for **Render Workflows**. If time allows on day 4, moving orchestration there takes a shot at the "Best use of Render Workflows" prize with little rework. Marked optional — must not slow the core build.

13 End-to-end system flow

How an assessment moves through the system, from click to PDF:

- 1 **Auth** — user signs in with Google (Auth.js). Session tied to their account.
- 2 **Select** — user picks a visa; the engine loads that visa's rule pack.
- 3 **Ingest** — links + résumé submitted. Server fetches GitHub + OpenAlex, parses the PDF.
- 4 **Extract** — Extractor agent turns everything into a structured evidence list (typed JSON).
- 5 **Retrieve** — each criterion paired with its governing policy text (fixed map) + relevant examples (semantic search).
- 6 **Deliberate** — per-criterion panel returns Met/Partial/Gap + reasoning + citations. Full panel only on borderline criteria.
- 7 **Decide** — code counts Met criteria, applies the N-of-M rule, computes any numeric checks. This is the verdict.
- 8 **Present** — scorecard renders: verdict, per-criterion cards, gap runway.
- 9 **Act & export** — Drafter agent writes the petition letter on request; user downloads the full PDF.

Data trust boundary: user-supplied links and files are treated as untrusted input — fetched server-side, parsed, and never used to drive actions beyond this assessment.

14 Scope for 5 days & feasibility

Build order — core loop first

PRIORITY	SCOPE
Must-have	Google auth · visa selector · ingest (links + résumé) · GitHub + OpenAlex fetch · Extractor · per-criterion panel · deterministic count · scorecard UI · PDF export · deploy — for O-1A
Should-have	EB-1A pack · gap runway · draft petition letter · Policy Watch feed · lean F-1/H-1B pathway (odds + STEM-OPT + bridge) · saved history
Stretch	O-1B / EB-1B / NIW packs · personalized Policy Watch alerts · BLS salary check · Render Workflows

Risks & how they're handled

Accuracy / hallucinated law Grounding + deterministic verdict; the model can't invent criteria or cite non-existent policy.	Latency (many agent calls) Panel only runs full on borderline criteria; parallelize criteria; Gemini Flash is fast.
Data source down Every critical source is user-provided or a stable free API; graceful fallback to résumé-only.	Scope creep Strict must/should/stretch split; nothing on the critical path depends on a stretch item.

15 Judging alignment & what's next

Against the four criteria (25% each)

CRITERION	HOW CALIBER SCORES
Impact	Replaces a \$2,000, week-long barrier with a free 60-second one, and connects the whole immigrant journey (F-1 → green card) for an underserved, high-need group.
Creativity & originality	Adversarial per-criterion panel + the RFE predictor + living Policy Watch: framings no existing tool ships.
Technical execution	Multi-agent orchestration, grounded retrieval, a real AI/deterministic split, live data APIs — a system, not a wrapper.
Feasibility	Every data source is available online today; honest about the one sampled dataset. It actually runs.

Prize fit

Best Immigration Solution (O-1 filing, \$8k) — the demo shows the exact thing the prize enables, judged by O-1 experts. **Best Overall** (\$2,500) — cleanest end-to-end product wins; polish + honesty is the play.

What's next after the hackathon

- Full BLS wage integration across all occupations.
- Deepen O-1B / EB-1B / NIW into first-class assessments.
- Evidence-collection workflow: turn a gap runway into tracked tasks.
- Optional human-lawyer handoff — export the packet straight to a partner firm.

A Appendix — the 8 O-1A criteria

An O-1A petition must demonstrate at least **three** of the following (per 8 CFR 214.2(o)(3)(iii)). This is the fixed skeleton the engine reasons against; each other visa swaps in its own list.

#	CRITERION	PRIMARY EVIDENCE SOURCE IN CALIBER
1	Nationally/internationally recognized awards	Résumé + user input
2	Membership requiring outstanding achievement	Résumé + user input
3	Published material about you	User-pasted press links
4	Judging the work of others	Résumé (peer review, panels)
5	Original contributions of major significance	GitHub + OpenAlex
6	Scholarly articles	OpenAlex
7	Critical role for distinguished organizations	Résumé + user input
8	High salary / remuneration	User input vs BLS data

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